# FINAL UI / UX
# PAGE LAYOUT SPECIFICATION

## AICTE Approval & Compliance Orchestration Platform

### Problem Statement 26262

> **Purpose**
> A build-ready page-by-page UI specification for the responsive web application. It defines layout format, navigation, tabs, tables, forms, actions, role permissions and key interaction states.

## Design decisions

- Web-first, desktop-optimized and responsive to tablet and mobile browsers.
- One shared application with role-based workspaces, navigation and server-side authorization.
- Common institution, application, EVC, obligation, document, notification and audit records.
- Configurable workflows and deadline rules; actual APH rules and permissions must be validated with AICTE.
- AI may assist with extraction and summaries, but official scrutiny, verification and decisions remain with authorized people.

## Document contents

1. Shared design system and global shell | 2. Role-based navigation | 3. Thirty page/screen specifications | 4. Reusable interaction and validation rules | 5. Responsive behavior | 6. MVP build order and acceptance checklist

*Basis: Problem Statement 26262 and the two supplied documents: AICTE Approval & Compliance Project Solution Blueprint (21 pages) and AICTE PS26262 Simple Explanation Guide (12 pages). This is a product design proposal, not an official AICTE UI standard.*

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## Other sections


- Part I - Shared design system and page format
- Part II - Role-based navigation and route map
- Part IV - Shared interaction, validation, security and accessibility standards
- Part V - Implementation order and acceptance checklist


## How to use this specification

Each screen definition includes its route, roles, purpose, layout order, tabs, table/form fields, primary actions and validation/permission rules. Screen specifications begin on a fresh page for easier handoff to the UI developer. Larger detail screens may continue onto a second document page.

# PART I - Shared design system and page format

## 1. Application frame

| Region | Desktop format | Behavior |
|---|---|---|
| Global header | Height 64 px; fixed or sticky; full content width. Left: application mark/name. Centre/right: global search, help, notifications, profile. | Keep consistent on all authenticated pages. Hide or compact secondary items on mobile. |
| Left sidebar | Width 240-256 px; vertical navigation; collapsible. Group related links and show active item clearly. | On tablet use collapsible sidebar; on mobile use a drawer or bottom nav for selected frequent actions. |
| Breadcrumbs | Below header, above page title; e.g., Applications > A-1042 > Workflow. | Only show when it improves orientation. Each parent link must respect permissions. |
| Page heading | Title, one-line purpose, status or scope when appropriate, right-aligned primary action. | Primary action is singular and context-sensitive; avoid multiple competing buttons. |
| Toolbar | Search, filter button, date range, saved views, sort, export if authorized. | Use filter chips for applied filters and provide a clear-all action. |
| Content grid | 12-column responsive grid; cards and panels on a 8 px spacing scale. | At desktop, use 2-3 columns where useful; tablet 2 columns; mobile 1 column. |
| Side context panel | Optional 280-340 px; owner, deadline, next action, related items. | Use only on detail/review pages where it reduces navigation. |
| Footer/status | Optional system state, last sync time or support links; no unnecessary global clutter. | Data freshness is shown on pages that rely on imported records. |

## 2. Visual tokens

| Token | Recommended specification | Usage |
|---|---|---|
| Primary navy | #102B46 | Header, major navigation, high-level headings. |
| Action blue | #1B527A | Primary buttons and focused links. |
| Accent teal | #087E8B | Section accents, active navigation and selected highlights. |
| Canvas | #F4F7FA | Overall background. |
| Surface | #FFFFFF | Cards, forms, tables and panels. |
| Border | #D5E0E8 | Separators and card outlines. |
| Success | #267A53 | Verified/completed states, paired with text. |
| Warning | #B87917 | Due soon and attention required. |
| Critical | #AC3E3E | Overdue, failed or rejected states. |

| Token | Recommended specification | Usage |
|---|---|---|
| Typography | Aptos / Arial or equivalent sans-serif | Readable hierarchy; body 14-16 px in UI; compact dense-table text no smaller than comfortable readable size. |
| Spacing | 8 px base scale | Consistent padding, table density and card gaps. |
| Controls | 8 px radius; 40-44 px minimum touch target | Inputs, buttons, filters and touch interactions. |

## 3. Standard page skeleton

HEADER (global search + notifications + profile)
LEFT SIDEBAR (role-based navigation)
BREADCRUMB
PAGE TITLE + DESCRIPTION + PRIMARY ACTION
FILTER / SEARCH TOOLBAR
KPI CARDS (only when meaningful)
MAIN WORK AREA (table, form, timeline, calendar or document review)
CONTEXT PANEL (owner, next action, deadline, related records)
PAGINATION / EMPTY STATE / ERROR STATE

## 4. Components shared across pages

| Component | Standard behavior |
|---|---|
| Status chip | Text and restrained color; never color-only. Keep terms consistent. |
| KPI card | Label, value, optional comparison/date; clicking filters the underlying records. |
| Data table | Server-side sort/filter/pagination; persistent filters where appropriate; clear empty/error/loading states. |
| Detail tabs | Keep tab order consistent; preserve the selected tab when returning to a record if feasible. |
| Timeline | Newest/oldest order chosen consistently; show actor, timestamp, action, reason and evidence link. |
| Confirm dialog | For consequential actions: summarize change, require reason when needed, confirm or cancel. |
| Document upload | File type/size checks, progress, failure recovery, preview, version tracking and permission checks. |
| Notification item | Event, why it fired, related record, due date, owner, delivery state and action link. |
| Audit note | Do not silently rewrite important events; record correction as a new authorized event. |
| Empty/error state | Explain why nothing is shown and offer a permitted next step; never fabricate sample values in production. |

## 5. Role and scope principles

- Institution representatives see only their institution and permitted application/obligation records.
- Regional and Bureau officers see records within their authorized scope and assignments.
- EVC users see assigned visit and report work; they do not automatically administer APH rules.
- Compliance reviewers see assigned obligations and related evidence within their scope.
- Management sees authorized consolidated summaries and only the detailed records their role permits.
- System administrators handle technical configuration, accounts and health. Their role does not automatically confer regulatory decision authority.
- Every read/write/download/export is authorized server-side; frontend visibility is not a security boundary.

> **Rule of truth**
>
> Where an existing AICTE system is authoritative for official application status or decisions, read/synchronize it through approved integration. Do not silently overwrite it. Show source and last synchronization time.

# PART II - Role-based navigation

| Role | Sidebar / main tabs | Scope and key restrictions |
|---|---|---|
| Institution representative | Institution Dashboard; My Applications; Action Required; Compliance; Documents & Renewals; Notifications; Profile; Help | View own records, respond to deficiencies, submit evidence and renewals. No access to internal-only notes or unrelated institutions. |
| Regional officer | Overview; My Tasks; Applications; EVC Operations; Compliance; Documents; Alerts & Escalations; Institutions; Regional Reports | Region-scoped oversight and permitted workflow actions. |
| Scrutiny / Bureau officer | My Tasks; Applications; Scrutiny Queue; Re-scrutiny; Documents; Workflow History; Alerts | Perform assigned scrutiny actions and authorized transitions. |
| EVC coordinator / committee member | EVC Dashboard; EVC Queue; Schedule; Upcoming Visits; Visit Records; Reports Pending; My Tasks | Manage assigned EVC lifecycle and evidence. |
| Compliance reviewer | Compliance Overview; My Review Queue; All Obligations; Evidence Submitted; Clarification Required; Documents; Overdue | Verify evidence and record authorized outcomes. |
| Senior management / competent authority | Management Dashboard; Decision Queue; Escalations; Regional Performance; Reports; permitted record search | Consolidated oversight and decisions reserved to the role. |
| System administrator | Administration; Users & Roles; Workflow/APH Rules; Deadline Policies; Reference Data; Integrations; Audit; System Health | Technical administration and configuration; regulatory decisions remain separately permissioned. |

## Suggested route map

| Route | Screen |
|---|---|
| /login | Login and authentication |
| /dashboard | Role-adaptive overview |
| /tasks | My Tasks |
| /applications | Application explorer |
| /applications/:id | Application detail |
| /institution/dashboard | Institution dashboard |
| /institution/applications/:id | Institution application workspace |
| /institution/compliance | Institution obligations and renewals |
| /evc | EVC workbench |
| /evc/:id | EVC detail and schedule |
| /compliance | Compliance reviewer workbench |

| Route | Screen |
|---|---|
| /compliance/submissions/:id | Evidence review detail |
| /documents | Document registry |
| /documents/:id | Document detail and versions |
| /alerts | Alerts and escalations |
| /institutions | Institution directory |
| /institutions/:id | Institution profile |
| /reports | Reports and analytics |
| /management | Management dashboard |
| /management/decisions | Decision and escalation queue |
| /admin/users | Users and roles |
| /admin/workflows | Workflow / APH configuration |
| /admin/policies | Deadline and escalation policies |
| /admin/reference-data | Reference data |
| /admin/integrations | Integrations and sync |
| /admin/audit | Audit log |
| /admin/health | System health |
| /notifications | Notification centre |
| /profile | User profile and preferences |
| /help | Help and guidance |

**Routing note**

Route names are proposed implementation conventions. They can be adjusted to the framework, but the role permissions and separation between application status, tasks, obligations, documents and alert status must be preserved.

# PART III - Page-by-page UI specifications

The layouts below specify the final content order and page format for each route. For screen implementation, use the shared application frame unless the page is unauthenticated or explicitly institution-facing.

## 01. Login and authentication

Route: /login | Users: All registered users

Purpose: Secure entry into a role-specific workspace.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Canvas | Full-height light neutral background. Center a 420-480 px login panel; keep brand and purpose visible. |
| Brand area | Logo/landmark mark, platform name, concise “Approval & Compliance” descriptor. |
| Form area | Username/email, password, show/hide password, approved MFA step if configured, sign-in button. |
| Support area | Forgot password / account recovery, configured support contact, privacy/security notice. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Username / email | Required; normalize whitespace; show generic auth error. |
| Password | Required; masked by default; never log in URL or logs. |
| MFA code (conditional) | Only when the configured identity provider requires it. |

### Primary actions

- Sign in.
- Start the approved recovery flow.

### States, permissions and UX rules

- Do not expose public staff-role registration unless policy explicitly allows it.
- Use secure session handling, rate limits and generic error messages.
- After login, route to the correct workspace based on verified role and scope.

# 02. Role-adaptive overview dashboard

Route: /dashboard | Users: Officials; content varies by role

Purpose: Show the work that needs attention first.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header strip | Page title, user scope/Region, reporting date range, last data refresh where applicable. |
| KPI row | 4-6 cards: active applications, pending tasks, overdue tasks, EVC due, compliance due and document expiry. |
| Middle grid | Left: Needs Attention list. Right: workload by stage, Region or task type. |
| Bottom grid | My Tasks table, upcoming deadlines, EVC schedule preview and recent activity. |
| Context panel | Optional: current owner, next action and why an item is overdue when an alert is selected. |

## Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| Overview | Role-relevant metrics and action items. |
| My Work | Tasks assigned to the current user. |
| Team / Region | Visible only to roles permitted to see aggregated scope. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| KPI card | Label, count, reporting scope and drill-down target. |
| Needs Attention row | Severity, record ID, institution, reason, owner and due date. |
| Workload chart | Metric definition, period and accessible table alternative. |
| Recent activity | Event, actor, timestamp, record and authorized link. |

## Primary actions

- Open a KPI to view the underlying filtered records.
- Open a task or alert.
- Change date range and permitted scope.

## States, permissions and UX rules

- Never show invented or hardcoded production metrics.
- Every displayed count must reconcile with defined database queries.
- Use skeleton loading, empty state, stale-data banner and API-error state.

# 03. My Tasks work queue

Route: /tasks | Users: All roles with assigned work

Purpose: Provide a single list of the actions assigned to the current user.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Title bar | My Tasks; count of open tasks; optional quick filter. |
| Filter row | Task type, status, due window, institution/application, priority and date. |
| Main region | Filterable table with owner, deadline, age and next action. |
| Detail interaction | Open task side panel or navigate to source record; do not create duplicated task details. |

## Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| My Open Tasks | Assigned tasks not completed. |
| Due Soon | Tasks inside configured warning period. |
| Overdue | Incomplete tasks whose applicable deadline passed. |
| Blocked | Tasks waiting on another authorized actor or prerequisite. |
| Completed | Recently completed tasks. |
| Reassigned | Delegated/reassigned items if the policy permits visibility. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Task | Short imperative action text. |
| Related record | Application/EVC/obligation/document ID. |
| Institution | Institution name + canonical ID. |
| Owner | Current responsible role/user. |
| Due / age | Due date, time remaining or overdue duration. |
| Status | Assigned, in progress, blocked, completed, cancelled with reason. |
| Next action | Clear action or blocker. |

### Primary actions

- Open source record.
- Perform permitted task action.
- Filter, sort and paginate.

### States, permissions and UX rules

- Complete/reassign actions require server-side permission checks.
- Task completion must update the appropriate workflow object and create an audit event.

## 04. Applications explorer

Route: /applications | Users: Institution staff within scope; officials

Purpose: Find and triage application records.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Page heading | Applications; optional Create/Import action for authorized roles. |
| Filter toolbar | Institution, application ID, type, Region, stage, status, owner, deadline and date range. |
| Applied filter chips | Visible active filters + Clear all. |
| Main table | Server-side searchable/sortable list with configurable columns. |
| Footer | Pagination, result count, saved views and permitted export. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Application ID | Official ID plus source ID where helpful. |
| Institution | Name and canonical identifier. |
| Type / route | Application category and applicable rule version. |
| Stage / status | Current stage and separate lifecycle status. |
| Owner / next action | Responsible person and outstanding task. |
| Deadline / ageing | Applicable deadline and calculated age. |
| Last updated | Timestamp plus source freshness if synchronized. |

### Primary actions

- Open application detail.
- Save/reuse a filter view.
- Create/import only if authorized.
- Export only if permitted; log sensitive exports.

### States, permissions and UX rules

- Default sort: overdue, then due soon, then other pending records.
- Filtering and pagination happen server-side.
- Do not let a list-row inline edit bypass workflow transitions.

# 05. Application detail workspace

Route: /applications/:id | Users: Authorized institution/official users according to record scope

Purpose: Show the application state, required next action and its complete history.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Identity header | Application ID, institution, application type, Region/source, lifecycle status and current stage. |
| Summary strip | Current owner, next action, applicable rule version, due date and age. |
| Primary action | One context-sensitive button such as Perform workflow action; secondary View history. |
| Main body | Tabs across application overview, workflow, scrutiny, submissions, EVC, compliance, communications and audit. |
| Right context panel | Outstanding blockers, deadlines, linked obligations, alerts and source freshness. |

### Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| Overview | Institution, type, stage, owner, next action, dates and blockers. |
| Workflow Timeline | Chronological stage transitions, returns, reassignment and actions. |
| Scrutiny & Observations | Checklist, observations, deficiency reasons, outcome and reviewer remarks. |
| Submissions & Documents | Current and earlier versions, supporting files and verification state. |
| EVC | Requirement, queue, schedule, visit and report states. |
| Compliance | Approval-linked obligations, deadlines, evidence and verification status. |
| Communications | Authorized notices, clarifications and replies. |
| Audit History | Detailed event history under strict permissions. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Current stage | Read from active workflow instance. |
| Rule version | Version that governs the instance. |
| Owner | Current task owner, not merely last actor. |
| Next action | Action from active task/transition definition. |
| Source ID | Original system and source record key. |
| Timeline event | Actor, action, timestamp, reason and evidence reference. |

### Primary actions

- Execute only permitted stage actions.
- Return for correction with reason and deficiencies where authorized.
- Assign/reassign where permitted.
- Open linked EVC, obligation, document or alert.

### States, permissions and UX rules

- Do not show all actions for all states.
- Official decisions remain with the competent authority.
- Preserve the prior state and return/resubmission history.

## 06. Institution dashboard

Route: /institution/dashboard | Users: Institution representative

Purpose: Give the institution a simple view of its applications, actions and upcoming obligations.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Institution header | Verified institution name and identifier; account/authorization status. |
| KPI cards | My applications, actions required, upcoming obligations, submissions under review, expiring documents. |
| Main grid | My Applications list beside Action Required panel. |
| Lower panels | Upcoming compliance deadlines, recent updates and document renewal reminders. |

### Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| Overview | Summary and actions. |
| My Applications | All permitted applications. |

| Tab / section | Contents and behavior |
|---|---|
| Action Required | Deficiencies, clarification requests, evidence due. |
| Compliance | Obligations and due dates. |
| Documents | Current and renewed files. |
| Notifications | Official messages visible to the institution. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Application | ID, type, visible status and last update. |
| Action required | Clear required action, due date and link. |
| Obligation | Condition summary, due date, evidence requirement and state. |
| Submission | Submitted at, version and review status. |

### Primary actions

- Open application.
- Respond to a correction request.
- Submit evidence or renewal.
- Open notification.

### States, permissions and UX rules

- Institution users must not see internal-only notes, other institutions or restricted decision records.
- Show “Submitted” and “Verified” separately.

## 07. Institution application workspace

Route: /institution/applications/:id | Users: Institution representative

Purpose: Let institution users view progress and respond to authorized requests.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Application reference, institution, visible status and current required action. |
| Progress strip | Completed/current/upcoming steps only where disclosure is permitted. |
| Action banner | Correction requested, due date, requirements and submission guidance. |
| Main tabs | Summary, Progress, Required Corrections, Submit/Resubmit, My Documents and Communications. |

### Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| Summary | Current visible status and application information. |
| Progress | Stage history suitable for institutional visibility. |
| Required Corrections | List each deficiency and required material. |
| Submit / Resubmit | File/field submission form with validation and confirmation. |
| My Documents | Uploaded versions and verification states. |
| Communications | Authorized correspondence and responses. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Correction item | Description, requirement source if shareable, due date and status. |
| Upload | Allowed type, size, required document category, file progress. |
| Response note | Required when policy calls for explanation. |
| Submission receipt | Timestamp, submitter and version reference. |

### Primary actions

- Save draft if permitted.
- Submit corrections.
- Download accessible documents.
- Respond to clarification.

### States, permissions and UX rules

- Keep previous submission versions.
- Warn before navigating away from unsaved form edits.
- Use resumable/retryable uploads where practical.

## 08. Institution compliance and renewals

Route: /institution/compliance | Users: Institution representative

Purpose: Show institutional obligations and allow authorized evidence submission.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Compliance & Renewals; institution scope. |
| Summary cards | Due soon, evidence submitted, awaiting verification and overdue. |

| Area / region | Final layout specification |
|---|---|
| Tabs | All Obligations, Due Soon, Submitted, Awaiting Verification, Overdue, Completed. |
| Main list | Obligation, related approval, required evidence, due date, status and action. |
| Detail drawer/page | Condition text, evidence checklist, prior submissions, reviewer status and communication. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Obligation ID | Unique reference. |
| Requirement | Plain-language summary with official source reference where permitted. |
| Related approval | Application/approval reference. |
| Due date | Applicable authorized deadline. |
| Required evidence | File/category checklist. |
| Status | Due, submitted, under review, clarification, compliant, overdue or configured outcome. |
| Action | View, submit, resubmit or respond. |

## Primary actions

- Open obligation details.
- Upload evidence and submit.
- Respond to clarification.
- View earlier submissions.

## States, permissions and UX rules

- A submission does not equal compliance.
- Do not allow institution users to edit official verification state or legal condition text.

# 09. EVC workbench

Route: /evc | Users: EVC coordinator, committee member and authorized officers

Purpose: Manage EVC queue, scheduling, completion and report follow-up.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| KPI row | Queue, scheduling due, upcoming visits, visit complete/report pending. |

| Area / region | Final layout specification |
|---|---|
| Filters | Region, application type, EVC state, coordinator, due window and new-institution flag where applicable. |
| Tabs | EVC Queue, Calendar/Schedule, Upcoming Visits, Visit Completed, Report Pending, Complete for Stage. |
| Main area | Switch between table and calendar for schedule-oriented views. |
| Side panel | Selected EVC owner, due date, blockers and next action. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Application / institution | Record identifiers. |
| EVC requirement | Required/not required and rule basis. |
| EVC state | Awaiting eligibility, queued, scheduled, visit completed, report pending, complete. |
| Coordinator | Assigned authorized owner. |
| Visit date | Confirmed date plus change history. |
| Deadline | Applicable deadline and age. |
| Next action | Schedule, record visit, upload report or review. |

## Primary actions

- Open EVC details.
- Schedule or reschedule where authorized.
- Record visit completion.
- Upload/review report if role allows.

## States, permissions and UX rules

- Scheduling, visit completion and report submission are separate events.
- Conflicts or incomplete prerequisites must be visible, not silently bypassed.

# 10. EVC detail and schedule

Route: /evc/:id | Users: Assigned EVC users and authorized coordinators

Purpose: Track one visit from eligibility to stage completion.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | EVC ID, application, institution, current EVC status and deadline. |
| Summary | Eligibility, owner, visit date, assigned participants and pending items. |

| Area / region | Final layout specification |
|---|---|
| Action area | Schedule/reschedule, record completion, submit report or record follow-up per permissions. |
| Tabs | Overview, Scheduling, Visit Record, Report & Review, History. |
| Right panel | Next action, reminders and blockers. |

### Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| Overview | Application link, requirement basis and EVC state. |
| Scheduling | Date, times, participants, updates and communications. |
| Visit Record | Completion event and permitted evidence. |
| Report & Review | Report submission, review state, required follow-up. |
| History | Chronological EVC events. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Visit date/time | Required for confirmed schedule; timezone rules defined by deployment. |
| Participants | Only authorized roster/roles. |
| Change reason | Required for reschedule/cancellation if policy says so. |
| Report reference | Secure file ID, submission timestamp, reviewer and state. |

### Primary actions

- Confirm or change schedule.
- Record visit completion.
- Record report receipt/review.

### States, permissions and UX rules

- Do not mark EVC complete while required stage actions remain incomplete.
- Any reschedule/cancel action creates a history event and updates reminders safely.

## 11. Compliance reviewer workbench

Route: /compliance | Users: Compliance reviewers and authorized managers

Purpose: Triage obligations and evidence that require official review.

### Page format and layout

| Area / region | Final layout specification |
|---|---|

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Obligation ID | Unique record key. |
| Institution / approval | Linked entities. |
| Condition | Short summary with source reference. |
| Due / submitted date | Store separately. |
| Reviewer | Current owner. |
| Evidence status | Not submitted, submitted, under verification, correction requested, verified, rejected. |
| Action age | Elapsed time in review. |

### Primary actions

- Open evidence review.
- Assign/reassign within scope.
- Send permitted clarification request.
- Review overdue and escalate under policy.

### States, permissions and UX rules

- Separate obligation status from alert status.
- Use explicit reason codes and review outcomes.

## 12. Evidence review detail

Route: /compliance/submissions/:id | Users: Assigned compliance reviewer

Purpose: Review evidence against a specific obligation and record an auditable outcome.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Institution, obligation ID, submission version, due date and current review status. |
| Desktop 3-panel layout | Left: condition/checklist. Centre: secure document viewer. Right: submission metadata and review form. |
| Mobile/tablet layout | Stack condition, document, then decision form; keep key actions fixed near form end. |
| History strip | Earlier submissions and prior clarification/rejection outcomes. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Condition checklist | Configured evidence requirements with completion state. |
| Document viewer | Page navigation, zoom, download only if permitted. |
| Review outcome | Verified, clarification required, rejected or other configured outcomes. |
| Reviewer remarks | Required for rejection/clarification where policy requires. |
| Evidence reference | Store document and page references for any AI-assistance note. |

### Primary actions

- Record verification outcome.
- Request clarification.
- Reject with reason.
- Return to queue.

### States, permissions and UX rules

- AI extraction is advisory and visually distinguishable from verified values.
- Only assigned/authorized reviewers may submit a verification decision.
- A new submission creates a new version; prior review history remains.

## 13. Document registry

Route: /documents | Users: Authorized institution-facing or official users by scope

Purpose: Find documents by institution, type, expiry, renewal and verification state.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Documents; upload/import action only for authorized users. |
| KPI strip | Expiring soon, expired, renewal submitted, verification pending. |
| Filter bar | Institution, document category, Region, expiry window, verification state and owner. |
| Tabs | All, Valid, Expiring Soon, Renewal Submitted, Awaiting Verification, Expired/Overdue, Rejected, Superseded. |
| Main table | Document record with state and next action. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Document ID | Stable document record ID. |

| Field / column | Format and validation |
|---|---|
| Institution | Canonical ID/name. |
| Category | Configured document type. |
| Valid until | Date plus confidence/source if extracted. |
| Verification | Verified, pending, rejected. |
| Renewal | Not submitted, submitted, verified. |
| Owner / action | Responsible reviewer and next action. |

### Primary actions

- Open document detail.
- Upload a new version where permitted.
- Filter/export under authorization.

### States, permissions and UX rules

- The file exists and the document is valid are separate facts.
- Keep expired and superseded versions accessible only to permitted users.

## 14. Document detail and version history

Route: /documents/:id | Users: Institution users and officials within scope

Purpose: Show document metadata, lifecycle, versions and verification evidence.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Document type, institution, current validity and verification status. |
| Summary strip | Issue date, valid-from, expiry, reviewer and related application/obligation. |
| Content area | Secure preview or file metadata; new version panel. |
| Tabs | Overview, Current Version, Version History, Extraction & Verification, Alerts, Audit History. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Document type | Configured category. |
| Issue / validity dates | Date fields with source and verification state. |
| Expiry date | Required only for categories where validity applies. |

| Field / column | Format and validation |
|---|---|
| Extraction data | Extracted value, confidence, source page/text and reviewer-confirmed value. |
| Version relation | Previous/superseded document ID. |
| Verification outcome | Status, reviewer, time and notes. |

### Primary actions

- Preview/download if authorized.
- Submit renewal/add version.
- Verify or request correction if reviewer.
- View related obligation/alert.

### States, permissions and UX rules

- Do not overwrite older files.
- Low-confidence extraction shows Review Required.
- Document downloads are independently authorized and logged where needed.

## 15. Alerts and escalations

Route: /alerts | Users: Officials, reviewers, management and administrators according to scope

Purpose: Track triggered alerts through delivery, acknowledgement, follow-up and closure.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| KPI row | Open, overdue, escalated, delivery failed. |
| Filter bar | Type, severity, Region, recipient, delivery status, acknowledgement and date. |
| Tabs | All, My Alerts, Upcoming, Overdue, Escalated, Delivery Failed, Acknowledged, Resolved. |
| Main table | Alert, related record, reason, deadline, recipient, delivery and next action. |
| Detail drawer | Trigger/policy, delivery attempts, acknowledgement, escalation and linked task. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Alert ID/type | Stable ID and category. |
| Related record | Application/EVC/obligation/document. |
| Trigger reason | Plain language and policy source. |

| Field / column | Format and validation |
|---|---|
| Recipient/channel | Authorized recipient and communication channel. |
| Delivery status | Queued, attempted, delivered, failed. |
| Acknowledgement | Actor and timestamp. |
| Resolution | Linked action completed/closed per policy. |

### Primary actions

- Open underlying record.
- Acknowledge.
- Retry/escalate only if authorized.
- Record follow-up.

### States, permissions and UX rules

- Acknowledgement does not close the underlying task.
- Notification retries use idempotency to avoid unintended duplicates.
- Sensitive contents must not leak into email previews.

## 16. Institution directory

Route: /institutions | Users: Authorized officers and management

Purpose: Find canonical institution records and see a permission-filtered summary.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Institutions; no public institution creation unless authorized. |
| Search toolbar | Institution ID, name/alias, Region and record source. |
| KPI row optional | Institutions with active applications, obligations due and documents expiring within scope. |
| Main table | Institution record and high-level activity summary. |
| Footer | Pagination, saved filters and authorized export. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Institution ID | Canonical identifier plus mapped source IDs. |
| Name / aliases | Official name and verified aliases. |
| Region | Reference value. |
| Applications | Count within scope. |
| Open obligations | Count with defined status basis. |

| Field / column | Format and validation |
|---|---|
| Document attention | Expiring/expired counts. |
| Last sync | Source and timestamp where relevant. |

### Primary actions

- Open institution profile.
- Resolve identity mapping/conflict if the role permits.

### States, permissions and UX rules

- Do not assume name similarity proves identity.
- Show source provenance and conflict states.

## 17. Institution profile

Route: /institutions/:id | Users: Authorized officials; institution user sees restricted own profile

Purpose: Present all relevant application/compliance records for one institution in a unified view.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Identity header | Institution name, canonical ID, Region, source identifiers and data freshness. |
| Summary cards | Applications in progress, overdue actions, obligations open, documents expiring. |
| Tabs | Overview, Applications, Compliance, Documents, EVC History, Communications, Activity History. |
| Main area | Authorized summary and related lists; tab tables retain their own filters. |
| Right panel | Next deadlines and unresolved tasks, when permission allows. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Institution identity | Canonical ID, official name, address/Region and source IDs. |
| Related application row | Application ID, type, current stage and owner. |
| Obligation row | Condition, deadline and verification status. |
| Document row | Type, valid-until, verified/renewal status. |
| EVC row | Visit date/state, report state. |

### Primary actions

- Open related application/obligation/document.
- Review unresolved items.
- Open identity mapping issue if authorized.

### States, permissions and UX rules

- Institution profile is a consolidated view, not a replacement for source records.
- Permission-filter each relation and document.

## 18. Reports and analytics

Route: /reports | Users: Regional/Bureau officers and management per scope

Purpose: Analyze workload, processing time, EVC, compliance, document validity and escalations.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Reports & Analytics; report selector and export permission. |
| Global filter bar | Period, Region, application type, stage, owner, status and deadline. |
| Report navigation | Overview, Application Processing, Ageing, Regional Performance, Workload, EVC, Compliance, Documents, Alerts, Audit Export. |
| Main report area | Charts accompanied by accessible data tables and definitions. |
| Drill-down area | Click a chart/category to open the records behind it. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Metric definition | Formula, included statuses and date basis. |
| Grouping | Region, stage, application category, owner or period. |
| Result table | Records underpinning the visual. |
| Export metadata | User, timestamp, filters and output format when logged. |

### Primary actions

- Apply filters.
- Drill down.
- Export authorized results.
- Save or reuse report view.

### States, permissions and UX rules

- Counts must reconcile to source records.
- Do not rank officials without agreed definitions/context.

- Exports require scope checks and appropriate logging.

# 19. Management dashboard

Route: /management | Users: Senior management and competent authority according to permission

Purpose: Provide consolidated oversight and highlight where intervention may be needed.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Reporting period, organizational scope and freshness indicators. |
| KPI row | Applications in progress, overdue tasks, EVC delayed, obligations outstanding, expired documents. |
| Main grid | Trend in pendency/processing; Region comparisons; stage bottleneck panel. |
| Attention queue | High-priority escalations and unresolved compliance items. |
| Bottom | Access links to detailed reports and permitted decision queue. |

## Tabs / sections

| Tab / section | Contents and behavior |
|---|---|
| Overview | Cross-region indicators and current risks. |
| Regional Performance | Compare authorized workload and ageing. |
| Bottlenecks | Stages with concentration of pending tasks. |
| EVC & Compliance | Delayed visits, report follow-up and outstanding obligations. |
| Escalations | Outstanding escalated work. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| KPI | Value, timeframe and definition. |
| Region row | Volume, overdue proportion, stage mix. |
| Bottleneck | Stage, queue size, age distribution and drill-down. |
| Escalation item | Reason, age, owner and next authorized action. |

## Primary actions

- Open source report/list.
- Open authorized decision queue.
- Filter by period/Region.

### States, permissions and UX rules

- Clearly mark data scope and date.
- Do not expose restricted personal or document content in aggregated views.
- Charts require table alternatives and accessible labels.

## 20. Decision and escalation queue

Route: /management/decisions | Users: Competent authority and roles delegated decision/escalation rights

Purpose: Give authorized decision-makers a controlled queue of matters that require their action.

### Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Decision Queue; scope and outstanding count. |
| Tabs | Awaiting Decision, Escalated Matters, Returned for Clarification, Decision Recorded, Closed. |
| Queue table | Application, institution, stage, time waiting, review material, deadline and escalation reason. |
| Decision detail | Source evidence, observations, prior actions, recommendations where applicable and decision form. |
| Outcome confirmation | Confirm the authorized outcome and required reasons before recording. |

### Fields / table columns

| Field / column | Format and validation |
|---|---|
| Decision item | Application ID, current stage and decision type. |
| Evidence bundle | Authorized docs, observations and source links. |
| Recommendation | If present, clearly separate from decision. |
| Outcome | Only configured, authorized options. |
| Reason/reference | Required where procedure dictates. |
| Decision actor/time | Automatically recorded. |

### Primary actions

- Open review bundle.
- Request clarification / return through authorized workflow.
- Record decision under permitted role.

### States, permissions and UX rules

- No AI-autonomous approval/rejection.
- Decision must validate role, current state and required evidence.
- Record actor, time, outcome and references in audit.

# 21. Users and roles administration

Route: /admin/users | Users: System administrator and authorized access managers

Purpose: Administer accounts, roles and authorized scopes.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Users & Roles; Add User action when permitted. |
| Tabs | Users, Roles & Permissions, Access Scope, Activation/Invitations if configured, Account Activity. |
| Search/filter | Name, user ID, role, Region, status and last login. |
| Main table | Users and assigned scope. |
| Detail page/drawer | Profile, roles/permissions, scope, account status and access history. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| User ID / email | Unique and verified identity value. |
| Role | Controlled role vocabulary. |
| Scope | Region/Bureau/institution/assignment as appropriate. |
| Status | Active, suspended, disabled, pending activation. |
| Reason | Required for sensitive access changes where policy requires. |

## Primary actions

- Create/disable account.
- Assign/revoke role.
- Change scope.
- Start approved recovery flow.

## States, permissions and UX rules

- Least privilege.
- Changes are audited.
- Admin permissions do not imply regulatory decision permissions.
- MFA/session controls depend on deployment identity provider.

# 22. Workflow and APH configuration

Route: /admin/workflows | Users: Authorized rule/configuration managers

Purpose: Configure, validate, test and publish versioned approval routes.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Workflow definitions; selected application type and version. |
| Tabs | Workflow Definitions, Application Types, Stage Catalogue, Transition Rules, Prerequisites/Checklists, Rule Versions, Draft Changes, Published Versions, History. |
| Builder workspace | Left: stage catalogue; centre: ordered stage/transition canvas; right: properties and rule conditions. |
| Properties form | Role, required evidence, prerequisites, deadline policy, permitted next stages and source rule reference. |
| Footer actions | Save draft, Validate, Simulate tests, Submit for authorization, Publish where permitted. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Rule/version | Unique version, effective dates, source reference. |
| Stage | Unique stage ID, display label and applicability. |
| Transition | From/to, prerequisites, authorized actor and condition. |
| Required evidence | Field/checklist requirements. |
| Deadline policy | Link to approved policy record. |
| Publication state | Draft, validated, pending authorization, published, retired. |

## Primary actions

- Create draft version.
- Run route simulation.
- Review validation errors.
- Publish through controlled approval.

## States, permissions and UX rules

- Never mutate a published version without creating an audited new version.
- Protect active instances from unreviewed route changes.
- Every route must have tests for valid and invalid transitions.

# 23. Deadline and escalation policies

Route: /admin/policies | Users: Authorized policy managers

Purpose: Configure approved deadline calculation, reminders and escalation policies.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Tabs | Deadline Policies, Reminder Windows, Escalation Rules, Working-Day/Holiday Calendars, Recipient Mapping, Change History. |
| Policy list | Name, related task type, effective period, status and version. |
| Policy editor | Trigger event, clock-start, calculation basis, warning windows, repeat behavior, exceptions and escalation recipients. |
| Test panel | Test sample start dates, working-day/holiday behavior, pauses and extensions before publication. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Policy ID/version | Unique version and effective dates. |
| Trigger/start event | Event that starts the clock. |
| Duration / due date | Authorized calculation and calendar convention. |
| Exceptions | Approved pause/extension rules only. |
| Reminder schedule | Lead time, repetition and channel. |
| Escalation | Threshold, recipient role and reason. |
| Source reference | Authorized policy reference. |

## Primary actions

- Draft/update rule.
- Test against cases.
- Submit/publish per governance.
- View change history.

## States, permissions and UX rules

- Prototype intervals are not official deadlines.
- Show the exact rule used for every calculated due date.
- Rule changes must be audited and effective-dated.

# 24. Reference data and document categories

Route: /admin/reference-data | Users: System administrators and authorized data managers

Purpose: Maintain centrally configured lists used across forms, filters and reports.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Tabs | Application Types, Regions, Departments/Roles, Stage Catalogue, Document Categories, Obligation Types, Notification Templates, Status Definitions. |
| Main table | Code, display label, active state, source/description and usage count. |
| Detail form | Create or edit permitted reference entry; effective date and deactivation status. |
| Change panel | Show records affected by a proposed category change. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Stable code | Unique machine-readable code; avoid changing keys in use. |
| Display label | User-facing name. |
| Description/source | Meaning and approved source where relevant. |
| Effective period | When the value may be used. |
| Active/inactive | Retire without deleting history. |
| Impact | Where the value is used. |

## Primary actions

- Add/edit authorized reference data.
- Retire unused values.
- Preview impact.

## States, permissions and UX rules

- Use controlled vocabularies.
- Do not delete values referenced by historical records.
- Require review for changes affecting active workflow rules.

# 25. Integration and synchronization

Route: /admin/integrations | Users: System administrator / integration operator

Purpose: Monitor source connectors, imports, conflicts and data freshness.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| KPI strip | Sources healthy, last sync failed, records rejected, conflicts open. |
| Tabs | Connected Sources, Sync History, Import Files, Validation Errors, Conflicts/Reconciliation, Data Freshness. |
| Source table | System, connector method, enabled state, last success, latency/status. |
| Job detail | Batch/event ID, rows read/accepted/rejected, error report and retry state. |
| Conflict panel | Source values, canonical value, field owner and authorized resolution action. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Source system | Named registered source. |
| Connector type | Approved API/event feed/file import. |
| Source record ID | Idempotent mapping key. |
| Last sync | Timestamp and state. |
| Validation result | Accepted/rejected with reason. |
| Conflict | Values, provenance and resolution. |
| Freshness | Current/stale according to configured threshold. |

## Primary actions

- Run authorized sync.
- Upload controlled import.
- Download validation report.
- Resolve conflict if permitted.

## States, permissions and UX rules

- Never show secrets.
- Repeated import must not duplicate records or side effects.
- Do not silently overwrite authoritative fields.
- Clearly indicate when data are stale or connector degraded.

# 26. Audit log

Route: /admin/audit | Users: Auditors and explicitly authorized officials

Purpose: Search and export the history of important record and system actions.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Audit Log; restricted-access indicator. |
| Filter bar | Actor, record type/ID, action, date range, Region, source system and outcome. |
| Main table | Timestamp, actor, action, record, outcome and reason summary. |
| Audit detail | Before/after values where appropriate, source reference, linked event IDs and evidence pointer. |
| Export | Authorized, filtered export with user/time logged. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Timestamp | Consistent timezone/display format. |
| Actor | User/service identity. |
| Action | Controlled action vocabulary. |
| Record | Record type and identifier. |
| Change details | Previous/new state or values where appropriate. |
| Reason/evidence | Required for sensitive transitions. |
| Event ID | Stable correlation identifier. |

## Primary actions

- Search/filter.
- Open event detail.
- Export authorized records.

## States, permissions and UX rules

- Normal users cannot edit/delete audit history.
- Restrict access to sensitive audit detail.
- Use new corrective events instead of silently rewriting history.

# 27. System health and operations

Route: /admin/health | Users: System administrators and operations personnel

Purpose: Show operational components that can compromise workflow, alerts or data freshness.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Status summary | API, database, worker, scheduler, notification provider, document storage and integrations. |
| KPI row | Queue backlog, failed notifications, sync failures, scheduler lag and error rate. |
| Tabs | Services, Background Jobs, Notification Delivery, Integrations, Backups/Recovery, Incident History. |
| Detailed table | Component, current state, last success, error count and latest failure. |
| Alert detail | Failure reason, affected records if safe, recovery action and incident history. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Component | Service/job/provider. |
| Health | Healthy, degraded, unavailable, unknown. |
| Last success | Timestamp. |
| Backlog/lag | Numeric operational measure when available. |
| Failure detail | Sanitized message, correlation ID and retry state. |
| Recovery | Last tested backup/restore or recovery run as applicable. |

## Primary actions

- Open logs/incident.
- Retry/restart only where safely supported and authorized.
- Acknowledge operational alert.

## States, permissions and UX rules

- Do not expose secrets or sensitive payloads in logs.
- Backups are not considered reliable until restore has been tested.
- Show degraded data source state to operational dashboards.

# 28. Notification centre

Route: /notifications | Users: All authenticated users

Purpose: Provide a searchable record of notifications relevant to the current user.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Notifications with unread count and mark-read actions. |
| Tabs | All, Unread, Deadlines, Workflow Updates, Compliance, Document Expiry; admin technical alerts only for authorized users. |
| Main list | Notification subject, reason, related ID, timestamp, delivery state and action link. |
| Detail panel | Full safe message, trigger date, due date and linked record. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Notification | Short descriptive title. |
| Reason | Why user received it. |
| Related record | Authorized link. |
| Timestamp | Generation/sent/received times where tracked. |
| State | Unread/read and delivery status where relevant. |

## Primary actions

- Open related record.
- Mark read.
- Manage non-mandatory preferences through profile.

## States, permissions and UX rules

- Do not include sensitive document contents in broad notification previews.
- Read/acknowledged state does not close the underlying task.

# 29. User profile and preferences

Route: /profile | Users: All authenticated users

Purpose: Let users view basic account details and edit only permitted preferences.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Profile summary | Name, user ID, role and scope; roles/scope read-only where centrally managed. |
| Tabs | Personal Details, Security, Notification Preferences, Sessions/Account Activity where supported. |
| Form layout | Single-column forms on mobile; two-column form only for logically paired fields on desktop. |
| Save behavior | Inline validation and explicit save confirmation. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Name/contact | Editable only if policy allows. |
| Role/scope | Read-only or request-change workflow. |
| Password/security | Use approved identity-provider flow, never store password in profile data. |
| Notification preferences | Optional channels/categories; mandatory alerts remain enabled. |
| Session list | Show and revoke sessions only if supported. |

## Primary actions

- Save permitted profile changes.
- Change password via approved flow.
- Adjust optional notification preferences.

## States, permissions and UX rules

- Do not permit users to self-grant roles or change authorization scope.
- Protect contact details and account history.

# 30. Help and guidance

Route: /help | Users: All users, content varies by role

Purpose: Help users understand statuses, perform tasks and contact support.

## Page format and layout

| Area / region | Final layout specification |
|---|---|
| Header | Help Centre; search help articles. |
| Left/content navigation | Getting Started, Application Workflow, EVC, Compliance, Documents, Alerts, Glossary, Support. |
| Article body | Short purpose, steps, screenshots/diagrams where useful, related links and last reviewed version. |
| Support panel | Configured support/contact method and current known service notices. |

## Fields / table columns

| Field / column | Format and validation |
|---|---|
| Article | Title, role scope, content version and reviewed date. |
| Search | Search title and content; results respect access. |
| Support request | Only if a real support workflow is configured. |

## Primary actions

- Search guidance.
- Open role-specific how-to.
- Contact configured support.

## States, permissions and UX rules

- Help content must match deployed route/rule labels.
- Do not present sample timelines as official AICTE timelines.
- Label prototype-only guidance clearly.

# PART IV - Shared interaction and validation standards

## 1. Standard form pattern

| Form area | Format |
|---|---|
| Page header | Title, purpose, breadcrumb and primary action. |
| Section grouping | Group fields by subject; avoid one very long undifferentiated form. |
| Required fields | Mark required inputs consistently; provide inline validation and error summary. |
| Dates | Use date picker plus readable display; show source/calculation basis for regulatory deadlines. |
| File fields | Show accepted formats, maximum size, upload state, scan/validation result and retry option. |
| Save/submit | Differentiate Save Draft from Submit; confirmation screen summarizes the resulting state. |
| Error handling | Preserve valid user inputs after validation failure; explain how to fix each issue. |
| Unsaved changes | Warn before leaving if material edits have not been saved. |

## 2. Workflow action confirmation format

When a user selects an action such as Complete Scrutiny, Return for Correction, Schedule EVC, Verify Evidence or Record Decision, open a context-specific confirmation form.

| Field | When shown / validation |
|---|---|
| Action | Pre-filled action name; cannot be changed to an unauthorized action. |
| Current state | Read-only current state from server; used to detect stale page actions. |
| Required reason | Mandatory for returns, rejection, rescheduling or other configured actions. |
| Checklist/evidence | Required before the configured transition can be accepted. |
| Next stage / recipient | Derived from rules and shown where useful; cannot bypass routing policy. |
| Confirm | Backend revalidates actor, scope, state, prerequisites and version before commit. |
| Result | Show success/failure and the new record state; do not imply success before server confirmation. |

## 3. Universal states

| Object | Suggested status vocabulary (configure to policy) |
|---|---|
| Application | Draft, Submitted, In Processing, Returned for Correction, Decision Recorded, Closed. |
| Workflow task | Not Started, Ready, Assigned, In Progress, Blocked, Completed, Cancelled with Reason. |
| Deadline | Not Yet Due, Due Soon, Due Today, Overdue, Paused by Authorized Rule, Completed. |
| EVC | Not Required, Awaiting Eligibility, Queued, Scheduled, Visit Completed, Report Pending, Complete for Stage. |
| Compliance obligation | Not Yet Due, Due Soon, Evidence Submitted, Verification Pending, Clarification Requested, Compliant, Overdue, Closed by Authorized Outcome. |
| Document | Submitted, Under Verification, Verified, Rejected, Expiring, Expired, Superseded by Verified Renewal. |
| Notification | Queued, Attempted, Delivered, Failed, Acknowledged, Linked Action Completed, Closed. |

## 4. Loading, empty, error and permission states

- Loading: skeletons for cards/tables; do not flash misleading zero counts while loading.
- Empty: state “No records match these filters” and provide Clear Filters where appropriate.
- No permission: explain that the record is not available to the account; do not disclose whether restricted data exists unnecessarily.
- Stale source: show the last successful sync and mark synchronized data as stale according to an agreed threshold.
- Integration error: provide a correlation/reference ID and show whether the affected record can still be safely viewed.
- Action failure: retain user inputs where safe, show the failed prerequisite and do not update UI state optimistically as final truth.
- Overdue state: show the rule/calculation basis and next owner/action when authorized.
- Mobile: stack panels and replace wide tables with cards; retain column labels and essential states.

## 5. Security and accessibility standards

- Enforce server-side authorization on every API call and document download.
- Use HTTPS, protected storage, file validation/scanning, secret management and protected backups.
- Log access to sensitive records and exports according to policy.
- Keep color paired with text; maintain sufficient contrast and keyboard navigation.
- Use accessible labels for forms, tables and charts; provide table alternatives for visualizations.
- Use role-sensitive notification content and do not expose confidential information in email previews.
- Test restoration from backups and simulate worker restart/integration failure.
- Keep official regulatory decisions with the competent authority.

# PART V - Implementation order and acceptance checklist

## Recommended implementation sequence

| Phase | Screens / capabilities | Exit condition |
|---|---|---|
| 0. Validate | Current portal capability inventory, APH/rule version, role matrix, deadline source, data ownership, integration permissions. | Process owner confirms what exists, what must be reused, and authoritative source for each field. |
| 1. Foundation | Login, Users/Roles, Institution/Applications records, navigation, basic audit. | Users see only permitted records; test data imports without duplicates. |
| 2. Workflow | Applications explorer/detail, task queue, workflow config for one route, return/re-scrutiny. | Valid transitions work; invalid and unauthorized transitions are rejected. |
| 3. Deadline/EVC | EVC workbench/detail, deadline policies, alert centre, background worker. | Due/overdue calculations and reminders survive retry/restart tests. |
| 4. Compliance/documents | Institution compliance, compliance review, document registry/detail and renewal versions. | Submitted evidence cannot become verified without authorized review. |
| 5. Dashboards/reporting | Role dashboards, reports and drill-downs, institution profile. | Counts reconcile to database and are permission-filtered. |
| 6. Integration/security | Integration monitor, audit log, system health, backup/restore and security checks. | Source conflict, stale-data, delivery failure and recovery scenarios pass. |

## Prototype acceptance checklist

- ☐ At least one configured route selects the correct stages based on application type and rule version.
- ☐ Return/re-scrutiny stores the reason, requested correction and submission version history.
- ☐ EVC queue, schedule, visit completion and report-pending states are distinct.
- ☐ An approval-linked obligation can be created, submitted, reviewed and closed with a complete history.
- ☐ A document renewal creates a new version and preserves the old file record.
- ☐ Deadline worker generates reminders, records failed delivery and retries without unintended duplicates.
- ☐ Alert acknowledgement does not close the underlying incomplete task.
- ☐ An unauthorized user cannot view/download restricted records or perform protected actions.
- ☐ Role dashboards show authorized scope only and their KPI values reconcile to underlying records.
- ☐ Integration or import failures are visible; stale source data and conflicts are not silently hidden.
- ☐ Audit events exist for important state changes, verification, rule changes, alert escalation and access-sensitive actions.
- ☐ Synthetic demo records are visibly labeled and never presented as live AICTE statistics.

> **Final recommendation**
>
> Deliver a responsive web application first. Build the central data model, versioned workflow, task ownership, deadline worker, EVC lifecycle, obligation registry, document versioning and audit history before investing heavily in decorative analytics or an autonomous AI layer.

## Important assumptions and boundaries

- This is a proposed UI/UX specification derived from the supplied Problem Statement 26262 and two project PDFs.
- It does not establish that any capability is absent from current AICTE systems. Existing portal capabilities and data access must be audited before duplicating functionality.
- APH versions, application routes, prescribed deadlines, role definitions, allowed transitions, retention rules and escalation recipients require confirmation from authorized AICTE sources/process owners.
- AI can assist with OCR, extraction, summaries and anomaly flags. It must not invent regulatory rules or replace scrutiny, inspection, verification or competent-authority decisions.

*Source basis: AICTE Approval & Compliance Project Solution Blueprint (user-supplied PDF, 21 pages) and AICTE PS26262 Simple Explanation Guide (user-supplied PDF, 12 pages).*